

Integrated Motor-Pump with a Spanwise-Varying Multi-Airfoil Rotor for Cooling of AI Datacenters and Edge Compute Nodes

Yuming Yan
Elmore Family School of
Electrical
and Computer Engineering
Purdue University
West Lafayette, USA
yan568@purdue.edu

Scott D. Sudhoff
Elmore Family School of
Electrical
and Computer Engineering
Purdue University
West Lafayette, USA
sudhoff@purdue.edu

Taehoon Chin
Elmore Family School of
Electrical
and Computer Engineering
Purdue University
West Lafayette, USA
thchin0726@purdue.edu

Woongkul Lee
Elmore Family School of
Electrical
and Computer Engineering
Purdue University
West Lafayette, USA
wklee@purdue.edu

Abstract—The growing adoption of artificial intelligence (AI) and data-intensive services in the 5G era has driven data centers toward higher computing capability and increased power density. The performance and reliability of high-performance computing systems are strongly constrained by their thermal management capability. This paper presents an integrated motor-pump with a spanwise-varying multi-airfoil rotor for cooling of AI datacenter and edge compute nodes. First, a spanwise-varying multi-airfoil rotor is designed that functions as the electromagnetic rotor and hydraulic blades simultaneously. Thus, such functional integration facilitates compact space and high-power density. Benefitting from essential axial flow, the bends and fittings are minimized reducing system-level hydraulic losses and increasing system efficiency. The design requirements for server-level cooling are specifically derived. The electromagnetic design and hydraulic design are investigated for server-level cooling. In addition, the spider struts are designed to reduce hydraulic resistance, and the systematic design is presented finally. The effect of stator skewing is discussed in terms of electromagnetic performance. The hydrodynamic performance and flow trajectories are investigated by computational fluid dynamics (CFD).

Keywords—integrated motor-pump, multi-airfoil, axial flow pump, multiphysics, hydrodynamics, AI data center, edge compute nodes.

I. INTRODUCTION

The continued growth of data-intensive applications and artificial intelligence has led to a sustained increase in energy consumption in datacenters and edge compute nodes over recent years, a trend that is expected to persist as computing workloads further expand across cloud and edge infrastructures [1]. Datacenters accounted for approximately 1-1.5% of global electricity consumption in 2022 and are projected to represent a significantly larger share over the next decade, driven by the rapid growth of AI and cloud computing workloads [2]. As system-level power density continues to increase and energy-efficiency requirements become more stringent, conventional air-cooling solutions are increasingly inadequate for modern AI datacenters, owing to their limited heat dissipation capability and inefficient space utilization [3]. Due to higher cooling capacity and space utilization, liquid cooling has attracted significant attention to enable direct-to-chip cooling in the emerging AI datacenters [4], [5]. Liquid cooling pumps are generally divided into positive-displacement and rotodynamic categories [6]. Positive-displacement pumps, such as gear and diaphragm pumps, deliver a fixed volume of fluid per cycle, whereas

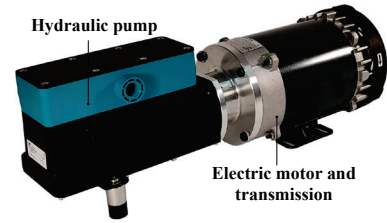


Fig. 1. Conventional pump with separate components for server-level cooling [9].

rotodynamic pumps rely on momentum exchange and are typically classified as centrifugal, axial-flow, or mixed-flow pumps [7]. The centrifugal pumps can provide a higher-pressure head in a single stage while axial-flow pumps offer advantages in terms of higher efficiency and a smaller cross-sectional area for a given flow rate [8].

Generally, liquid cooling pumps operate based on a high-speed motor, mechanical gear and hydraulic pump, as shown in Fig. 1 [9]. The electric motor and the hydraulic pump are separate components. As a result, more space is required to accommodate multiple sub-systems. Additionally, more bends and fittings are required that produce considerable system-level hydraulic losses and degrade the system efficiency and reliability [10]. Furthermore, it is inefficient to achieve multi-stage compression when a higher-pressure rise is required. To improve the torque/power densities of such high-speed motor and mechanical gearset combination, direct drive machines have received wide attention in recent years which eliminates the need of mechanical gear [11]–[13]. However, conventional direct drive machines mainly feature isotropic structure in axial direction. In other words, additional hydraulic pumps are still required to circulate the cooling loop using conventional direct drive machines. In the past decade, the functional integration among electric motors, motor drives and power electronics has become increasingly popular [14], [15]. This method has been recognized as a promising solution to facilitate next-generation energy conversion systems towards higher power density, higher efficiency, and higher reliability while reducing system cost [16], [17]. For instance, it has been reported that an airfoil-shaped rotor can be used for electric motor self-cooling [18]. However, it used simple single-airfoil profile for the rotor pole which essentially limits its aerodynamic performance.

This paper proposed an integrated motor-pump for cooling of AI datacenters and edge compute nodes. Unlike conventional pumps, a spanwise-varying multi-airfoil rotor is designed that simultaneously carry magnetic flux and

compress the inflow liquid, enabling direct conversion of electrical input power into compressed fluid power. In this way, the designed rotor functions as an electromagnetic rotor and hydraulic pump simultaneously. Furthermore, the spider struts are designed to reduce hydraulic resistance. A systematic hydraulic design based on these ideas is presented in this paper.

II. DESIGN REQUIREMENTS AND CONFIGURATIONS

A. Design Requirements of Server-Level Cooling

The integrated pumps should meet multifaceted requirements for cooling of AI datacenter and edge compute nodes. They should provide sufficient electromagnetic torque for hydrodynamic loading and overcome viscous and profile drag on the airfoil surfaces. Meanwhile, they should be compact to work in a limited volume of cooling tunnel. Furthermore, the integrated pump should meet the hydrodynamic requirements to guide the flow trajectories of coolant efficiently. An overview of our compact integrated pump is shown in Fig. 2. The mechanical power delivered by the rotating shaft can be formulated by

$$P_{shaft} = T * \omega \quad (1)$$

where T and ω are delivered torque and rotation speed of the shaft.

The useful power transferred from the rotor to the fluid can be calculated by [7]

$$P_{hyd} = \rho g Q H \quad (2)$$

where ρ is the fluid density, Q is volumetric flow rate, H is hydraulic head.

The overall hydraulic efficiency of the pump can be established as

$$\eta = \frac{P_{hyd}}{P_{shaft}} \quad (3)$$

According to the power conversion process in the hydraulic pump, the main design requirements can be obtained as summarized in Table I.

B. Configurations of Proposed Integrated Motor-Pump

Fig. 2 shows the configuration of the proposed integrated motor-pump, which features axial flow pump in essence. It contains three main parts, i.e., the multi-airfoil rotor, stator and winding, cooling tunnel and mechanical struts. The rotor is designed integrating electromagnetic benefits of salient rotor and hydrodynamic benefits of airfoil shapes into one unit. The stator leveraged that of flux switching machine comprising modular stator core, flux-concentrated PM and double-layer concentrated armature winding. The cooling tunnel and mechanical struts are designed to minimize the hydrodynamic resistance along coolant flow trajectories. With this unique configuration, the proposed integrated motor-pump can offer the following potential advantages.

1) The straight-through axial configuration enables a compact design that integrates naturally into tubular systems.

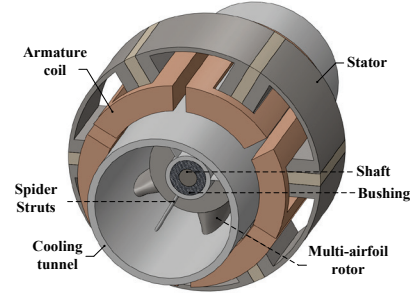


Fig. 2. Systematic design of proposed integrated motor-pump with spanwise-varying multi-airfoil rotor.

TABLE I
DESIGN REQUIREMENTS FOR THE INTEGRATED MOTOR-PUMP

Parameters	Initial design	Unit
Speed	6000	rpm
Input voltage	12	V
Outer diameter of stator	28	mm
Airgap length	1	mm
Active length	10	mm
Outer rotor diameter	14	mm
Flow rate	120	L/h
Head	2	m
Trailing edge thickness	≥ 0.3	mm
Tip clearance	≥ 0.1	mm
THD	≤ 15	%
Fluid	Water	/

2) The inline layout minimizes bends and fittings, reducing system-level hydraulic losses and increasing system efficiency.

3) The ducted geometry supports axial staging and straightforward integration of stators for performance enhancement.

III. ELECTROMAGNETIC DESIGN

The electromagnetic design aspects of the integrated motor-pump are discussed in this section.

A. Motor Topology

Among various motor topologies, the flux-switching permanent magnet (FSPM) machine is distinguished by its bipolar flux linkage and stator-mounted permanent magnets. In such way, the FSPM machine offers a robust salient rotor with enhanced reliability under high-speed operation. Moreover, the heat dissipation capacity of stator mounted magnets is enhanced which reduces the demagnetization risks. Meanwhile, the heat dissipation for the rotor is essentially enhanced due to hydraulic operation. Consequently, the FSPM machine is well suited for high-speed and integrated applications and is therefore investigated as the electromagnetic core of the proposed integrated pump system.

B. Consideration on Slot Pole Combinations

The appropriate combinations of stator slot and rotor pole number in FSPM machines are governed by [19]

$$N_s = k_1 m \quad k_1 = 1, 2, \dots \quad (4)$$

$$N_r = N_s \pm k_2 \quad k_2 = 1, 2, \dots \quad (5)$$

TABLE II
SPANWISE-VARYING MULTI-AIRFOIL ROTOR DESIGN

Section	Radius r (mm)	Solidity σ	Chord c (mm)	Thickness ratio t/c	Thickness t (mm)	Airfoil
Root	5.00	1.5	9.42	0.28	2.64	NACA 4428
Middle	6.00	1.225	9.24	0.24	2.22	NACA 4424
0.9R	6.75	1.125	9.54	0.20	1.91	NACA 4420
Tip	7.50	1.0625	10.01	0.20	2.00	NACA 4420

where N_s and N_r are the number of stator slot and rotor pole, respectively, and m is the number of phases. k_1 is an integer number when m is even and an even number when m is odd.

Assuming sinusoidal flux linkage, back EMF, and current waveforms, the electromagnetic torque is given by

$$T = \frac{3}{2} N_r \psi_M I_q = \frac{3}{2} N_r N_a \phi_M I_q = k_T I_q \quad (6)$$

IV. HYDRODYNAMICS DESIGN

In the integrated pump, the salient rotor poles of the FSPM machine are shaped into airfoils to provide pumping function. The motor torque production capability is related to the thickness of the rotor pole. The pumping efficiency is also dependent on the thickness of the airfoil profile. Therefore, the rotor pole thickness serves as a key parameter in designing the integrated pump. From the perspective of electric motor, the rotor pole thickness should not be too small to avoid saturation. However, in order to obtain high pumping performance, the blade thickness cannot be too large. Therefore, the blade thickness tradeoffs must be considered.

At the same time, the rotor pole height should not be too large to shorten the magnetic circuit. However, in order to obtain high pumping performance, the rotor pole height cannot be too small. Therefore, the rotor pole height tradeoffs must also be considered during rotor design.

The blade profile was selected from the National Advisory Committee for Aeronautics (NACA) in Fig. 3, focusing on those with a large thickness. NACA44 series blades are well suited due to their favorable lift characteristics and stable flow behavior under low Reynolds number conditions typical of small-scale axial-flow rotors [7]. To enhance pumping performance and align structural requirements, different airfoil profiles are arranged at different radial locations resulting in multi-airfoil rotor blades. The detailed profiles and corresponding characteristics are summarized in Table II.

Fig. 3 shows the blade profiles at different radial locations from rotor root to rotor tip. At the blade root, a relatively thick airfoil, i.e., NACA 4428, was selected to provide enhanced structural stiffness and mechanical robustness, which are critical in the high-stress region near the hub. The increased thickness also mitigates root bending deformation and facilitates reliable integration with the hub while maintaining sufficient camber for flow turning at low Reynolds numbers. In the inner-mid span region, the thickness ratio is moderately reduced, i.e., NACA 4424, to balance structural integrity with improved hydraulic efficiency. This profile provides adequate flow turning capability while reducing blockage and viscous losses compared to the root section, resulting in a smoother spanwise loading transition. Toward the outer span, where the local blade speed increases and structural loads decrease, a

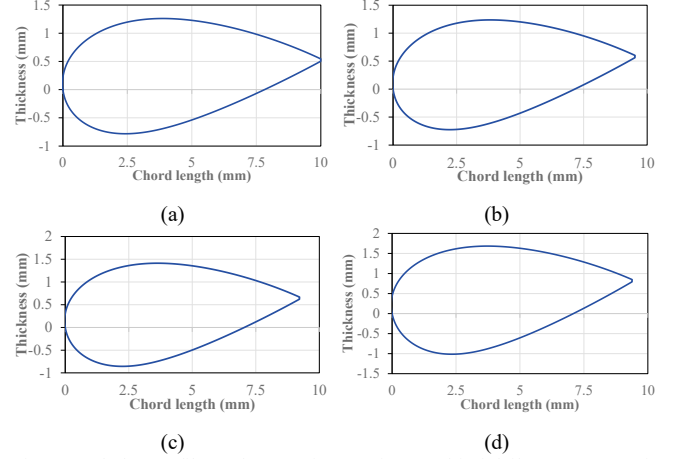


Fig. 3. Blade profiles of spanwise-varying multi-airfoil rotor. (a) Tip NACA4420 blade profile. (b) 0.9R NACA4420 blade profile. (c) Middle NACA4424 blade profile. (d) Root NACA4428 blade profile.

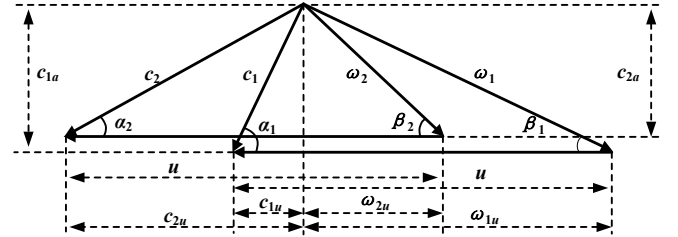


Fig. 4. Velocity triangle in an airfoil-shaped rotor (subscript 1 denotes the inlet, subscript 2 denotes the outlet).

thinner airfoil, i.e., NACA 4420, is adopted. The reduced thickness alleviates viscous drag and limits flow blockage, improving aerodynamic efficiency while maintaining sufficient camber for controlled flow deflection. At the blade tip, the same thinner profile is retained to minimize tip losses and reduce the strength of tip leakage vortices. The thinner section also limits pressure gradients near the tip clearance, contributing to smoother interaction with the surrounding duct and improved overall flow uniformity.

Fig. 4 illustrates the elemental velocity triangles at the inlet and outlet of the rotor blade section. The absolute velocity c , the relative velocity w , and the peripheral velocity u satisfy

$$c = u + w. \quad (7)$$

Under the assumption of negligible pre-swirl at the inlet, the circumferential component c_{1w} is small, and the inlet flow is predominantly axial. As the flow passes through the rotor, mechanical work is transferred from the rotating blade to the fluid, resulting in a change in the circumferential component of the absolute velocity. At the outlet, the absolute velocity c_2 exhibits an increased tangential component c_{2w} , which is directly responsible for the rise in fluid angular momentum and pressure according to the Euler turbomachinery equation

[7]. The relative outlet velocity ω_2 reflects the blade loading and governs the diffusion process within the blade passage.

The cross-section area of the rotor and axial velocity of the fluid can be calculated as

$$A_a = \frac{\pi}{4} (D_{ip}^2 - D_{hub}^2), \quad (8)$$

$$V_a = \frac{Q}{A_a}. \quad (9)$$

The circumferential velocity at different radial locations can be calculated

$$\omega = 2\pi n / 60, \quad (10)$$

$$U(r) = \omega r. \quad (11)$$

The solidity of the airfoil shapes is defined as [7]

$$\sigma(r) = \frac{Bc(r)}{2\pi r}, \quad (12)$$

where B is the number of blades, $c(r)$ is the length of chord;

The inlet metal corner can be established as [7]

$$\tan \beta_i(r) = \frac{V_a}{U(r)}. \quad (13)$$

V. MECHANICAL DESIGN AND MODULAR ROTOR

To consider hydrodynamic performance, manufacturability, and system integration simultaneously, modular rotor architecture is specifically designed for the integrated rotor as shown in Fig. 5. The rotor is decomposed into a laminated rotor core that provides high output torque and structural rigidity, and a set of detachable soft outer airfoil jackets. This enables different airfoil profiles, camber distributions, or materials to be selectively implemented without redesigning the core rotor. Moreover, this approach facilitates rapid iteration of blade shapes to improve flow guidance, reduce separation, and enhance efficiency, while also offering advantages in assembly, maintenance, and potential material compliance for noise and vibration mitigation.

Fig. 6 illustrates the developed tunnel configurations with multi-airfoil rotor. It consists of a 3D-printed PLA tube housing the multi-airfoil rotor supported by a steel shaft and a low-friction PEEK bushing that enables stable rotation with minimized mechanical losses. To condition the inflow and suppress swirl-induced losses, aerodynamically profiled spider struts based on symmetric NACA0012 airfoil are implemented at both the inlet and outlet. The inlet strut ($\theta_a = 0^\circ$) primarily serves as structural support while minimally disturbing the incoming flow, whereas the outlet strut ($\theta_a = 10^\circ$) functions as a flow rectifier to recover axial momentum and reduce downstream recirculation. This design provides a

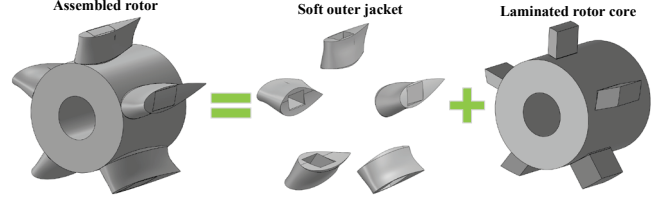


Fig. 5. Modular design scheme for multi-airfoil rotor.

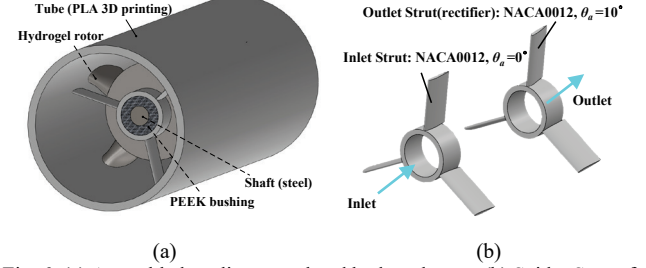


Fig. 6. (a) Assembled cooling tunnel and hydrogel rotor. (b) Spider Struts for minimized hydrodynamic resistance.

TABLE III
MAIN GEOMETRIC PARAMETERS OF THE SYSTEMATIC DESIGN

Parameters	Skewed	Straight	Unit
Rotor pole/blades	5		/
Stator slot	6		/
Inner rotor diameter	2		mm
Outer rotor diameter	15		mm
Current	1.1		A
Armature turns	57		/
Stator tooth width	1.5		mm
Magnet thickness	1.4		mm
Air gap length	1		mm
Tunnel thickness	0.8		mm
Outer tunnel diameter	16.8		mm
Inner tunnel diameter	15.2		mm
Bushing length	2.5		mm
Bushing pitch	12.5		mm
Spider strut base thickness	1		mm
Spider strut base length	2.5		mm
Shaft diameter	2		mm
Shaft	Steel		/
Tunnel	PLA		/
Bushing	PEEK		/
Rotor soft jacket	Hydrogel		/
Stator skew angle	4	0	°
Magnet skew angle	4	0	°

compact, manufacturable platform that balances structural integrity, low resistance, and effective flow management for small-scale cooling systems.

VI. MULTIPHYSICS PERFORMANCES EVALUATION

In this section, the electromagnetic characteristics of the multi-airfoil rotor are investigated by the FE analysis (FEA). To consider the skew effect of multi-airfoil rotor, the counterpart with skewed stator of the same skew angle is compared. The flow trajectories of the rotor are simulated and investigated by the CFD analysis. The main geometric parameters of the integrated pump are listed in Table III.

A. Electromagnetic Characteristics

Fig. 7 shows the cogging torque waveforms of the two machines at rated speed. It can be observed the peak-to-peak

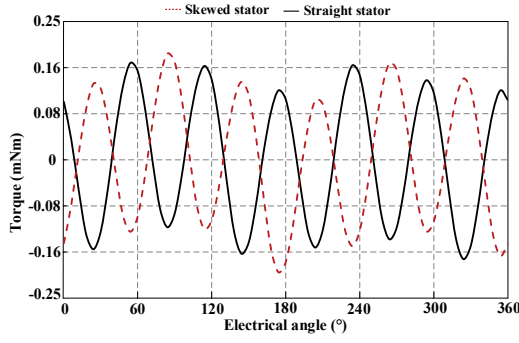


Fig. 7. Cogging torque waveforms of multi-airfoil rotor with skewed and straight stator.

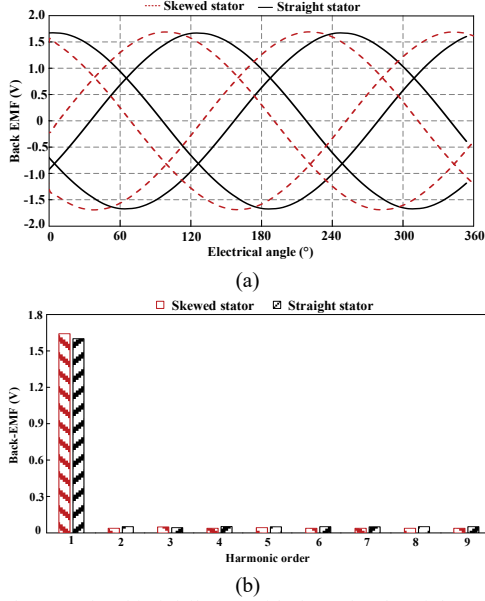


Fig. 8. Back EMF of multi-airfoil rotor with skewed and straight stator at rated speed. (a) Waveforms. (b) Spectra.

value of cogging torque in straight stator is 0.34 mNm while it is 0.36 mNm in skewed stator counterpart. Thus, the multi-airfoil rotor with straight stator exhibits lower cogging torque than its counterpart with skewed stator.

The back EMF waveforms and corresponding harmonic spectra are depicted in Fig. 8. It can be found that the fundamental components of the skewed stator and straight stator are 1.69 V and 1.67 V, respectively. The total harmonic distortion (THD) of skewed stator and straight stator is 4.76% and 6.22%, respectively. Thus, the skew of stator can effectively reduce THD of no load back EMF. Meanwhile, both skewed stator and straight stator can deliver comparable and sinusoidal back EMF waveforms.

Fig. 9 shows the magnetic flux density distributions of two machines under both no load and load operating conditions. It can be found local saturation mainly occurred around the axial extremities of the rotor pole due to designed multi-airfoil shapes. Thus, it is beneficial to design the rotor stack length to be relatively larger than that of the stator. Overall, the level of magnetic saturation is within a satisfactory range for the proposed design.

The steady output torque waveforms of two machines at rated speed and rated current are shown in Fig. 10. It can be found the average torque of the skewed stator and straight stator are 4.36 and 4.32 Nm, respectively. The corresponding

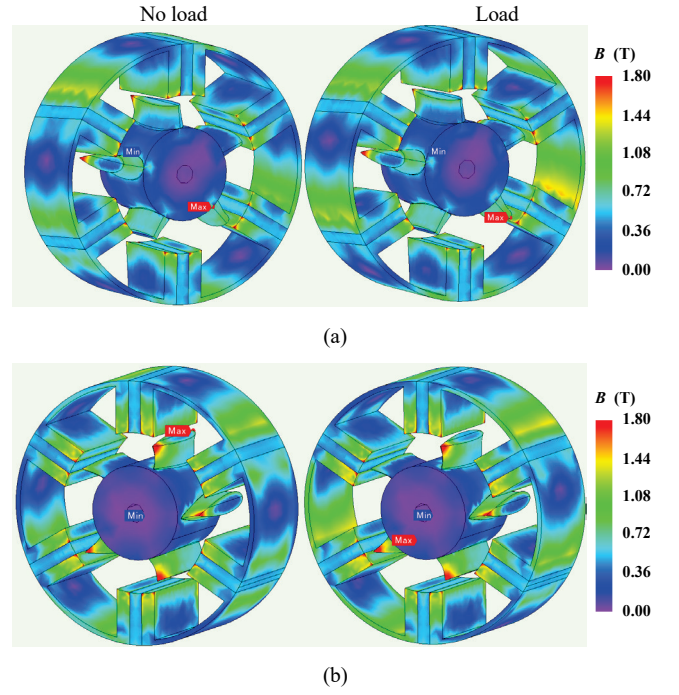


Fig. 9. Flux density distributions under no load and load conditions. (a) Skewed stator. (b) Straight stator.

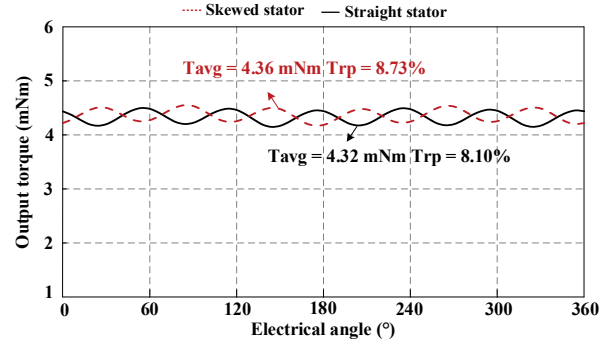


Fig. 10. Steady state waveforms of multi-airfoil rotor with skewed and straight stator.

TABLE IV
ELECTROMAGNETIC PERFORMANCES

Parameters	Straight	Skewed	Unit
Output power	2.71	2.74	W
Average torque	4.32	4.36	mNm
Torque ripple	8.10	8.73	%
Cogging torque	0.17	0.18	mNm
Air-gap flux density	0.37	0.33	T
Back-EMF	1.67	1.69	V
THD	6.22	4.76	%
Power density (active)	440.6	445	W/L

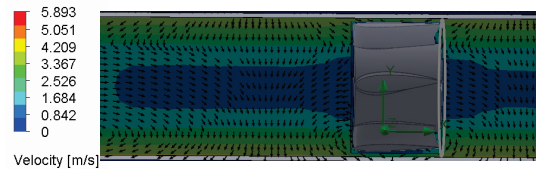


Fig. 11. Flow trajectories in the multi-airfoil rotor calculated by CFD at 6000 r/min.

torque ripples of the two machines are 8.73% and 8.10%, respectively. Therefore, both skewed stator and straight stator can produce comparable electromagnetic torque with quite

low torque ripple. The electromagnetic performances of two designs are summarized in Table IV.

B. Hydrodynamic Performances

Fig. 11 illustrates the CFD-predicted flow trajectories and velocity field of the proposed integrated motor-pump operating at 6000 r/min. The results show a clearly established net axial flow through the cooling tunnel, with fluid being effectively entrained by the rotating airfoil blades and accelerated downstream. The velocity contours and streamline patterns indicate smooth flow guidance across the rotor region, without large-scale flow separation or stagnant zones, confirming that the multi-airfoil rotor designs are well matched to the local velocity triangles. Moderate recirculation and shear near the wall are observed, which are characteristic of confined axial-flow devices at low Reynolds numbers, but do not dominate the core flow. Overall, it is demonstrated that the proposed design converts rotational motion into sustained axial momentum, thereby validating the hydrodynamic design.

VII. CONCLUSION AND FUTURE WORK

This paper proposed an integrated motor-pump with a spanwise-varying multi-airfoil rotor for cooling of AI datacenters and edge compute nodes. The specific design requirements for server-level cooling are revealed. The appropriate motor topology and slot pole combinations are elaborated. The hydrodynamic design of the multi-airfoil rotor is revealed. The modular design scheme for multi-airfoil rotor, spider struts for minimized hydrodynamic resistance and systematic design are presented. The electromagnetic and hydrodynamic performances are investigated by FEA and CFD. According to simulation results, it is demonstrated the proposed design can effectively generate electromagnetic torque and guide fluid flow simultaneously. Future work involves multi-physics optimization and benchtop experiments to validate the proposed systematic design.

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